Should the store offer 12 interest-free installments on Cyber Monday and Black Friday?

E-commerce analytics case study · Lucas Piccoli · **All data is simulated** (fictional store, seed 42)

> **Recommendation: keep 6 installments.** Offering 12 lifts sales per session by 10.6%, but cuts contribution after ads by 35%. Over a 3-day event, that is about **ARS 1.8M given up** (95% CI: −0.4M to +4.0M).

The decision

In Argentina, interest-free installments ("cuotas sin interés") are paid for by the merchant: roughly 5% of net sales for 3 installments, 10% for 6 and 20% for 12. The question is whether offering up to 12 (variant B) instead of up to 6 (variant A) raises conversion and basket size enough to pay for that extra cost. Sessions during both events were split 50/50 between the two variants (~55,000 sessions each).

A/B test results

Metric	A: up to 6	B: up to 12	B vs A	95% CI of the difference	p-value
Conversion rate	4.67%	4.82%	+3.2%	−0.10 to +0.40 pp	0.245
Average order value (ARS)	67,729	72,987	+7.8%	+2,068 to +8,448	0.001
Sales per 1,000 sessions (ARS)	2.99M	3.31M	+10.6%	+88K to +537K	bootstrap
Financing cost / net sales	5.0%	7.0%	+1.9 pp		
Contribution after ads per 1,000 sessions (ARS)	**95,028**	**61,719**	−35.1%	−73,687 to +7,004	bootstrap

Contribution = net sales − cost of goods − payment fees − installment financing cost − net shipping. "After ads" also subtracts ad spend. Normalising per 1,000 sessions captures conversion and basket size together, on equal traffic. Bootstrap: 10,000 resamples at session level.

Why the recommendation holds, even though −35% is not significant on its own

The contribution interval barely includes zero, so the test alone does not prove B earns less (probability that B beats A: ~5%). Three facts support the decision:

- **B needs far more conversion than it shows.** To match A's contribution it needs +12.4% conversion. It reached +3.2%, and even the top of its interval is +8.6%.
- **The test has enough power for the effect that matters.** It had 19% power to detect +3% (that would need 6.6× the sample), but 99% power to detect the +12.4% break-even.
- **Both events point the same way.** B is lower on Black Friday (61,987 vs 107,134) and on Cyber Monday (61,442 vs 82,547).

With returned orders refunded (a pessimistic scenario), contribution falls to 26,920 for A and **−5,023 for B**: B loses money after ads.

What would change the recommendation

- **A cheaper 12-installment plan:** at the observed conversion, B breaks even if its financing cost falls from 20% to 13.1% of sales. That is the first thing to negotiate with the payment provider.
- **A confirmed conversion lift of at least +12.4%** in a new test, detectable with about 22,000 sessions per variant (less than one event).

Beyond the test: what the rest of the dashboard shows

Promotions and profitability. Contribution margin drops from ~26% on regular days to ~12% during events, as discounts rise. Against a same-weekday baseline, both events have a negative promo ROI (−14.5% Cyber Monday, −13.6% Black Friday), with either variant.

Acquisition. Paid search returns 3.6× (3.2× after cancellations and returns) and paid social 2.4× (2.2×). But observed LTV is below CAC in both: 19,164 vs 23,395 for search, 19,293 vs 33,046 for social. LTV covers 4 months at most, so it is a floor.

Operations. Late deliveries rise from ~11% on regular days to 28.6% on Cyber Monday and 30.5% on Black Friday (2.6×). Stockouts double (~3% to ~7.5%), and late delivery becomes the top reason customers contact support.

Data quality. 12 of 13 reconciliation checks pass. The one that fails is documented: 1,809 order lines (2.97%) were sold while marked out of stock. Excluding those orders, A stays at 57,293 and B at 16,571 per 1,000 sessions, so the conclusion does not change.

Method

- **Statistics:** two-proportion z-test for conversion, Welch's t-test for order value, session-level bootstrap for sales and contribution per 1,000 sessions, break-even conversion, power analysis and minimum detectable effect.
- **Metric design:** the decision metric measures money, not volume. Every rate is computed as SUM(numerator) / SUM(denominator), never as an average of rates. Two contribution definitions (returns kept vs refunded) are shown side by side.
- **Data modelling:** 7 source tables loaded into BigQuery with explicit schemas, 19 SQL views (A/B test, funnel and acquisition, promotions, cohorts and LTV, operations, data quality).
- **Validation:** the same views run in DuckDB and match 22 control values.
- **Dashboard:** 6 pages in Looker Studio, each with a title that states its finding. A Power BI star schema with DAX measures is also documented.

Stack: Python (pandas, NumPy, SciPy), SQL, BigQuery, DuckDB, Looker Studio, Power BI (DAX).

Limits

- The data is simulated. The effect of installments on conversion and basket size is an assumption of the data generator: this project demonstrates the **analysis method**, not a finding about real shoppers.
- The data does not record post-event behaviour, so it cannot tell whether 12-installment buyers return more or less.
- Sales lost to stockouts are not recorded and cannot be estimated.

Links

Dashboard:	datastudio.google.com/reporting/a4715b0f-c25e-4335-94e4-1e2fbb5f4cf6
Code and documentation:	REPO_URL